Fab Automation and Software Engineer

TSMC · Austin, TX

Job ID (this archive)	MFG-028
Employer	TSMC
Location	Austin, TX
Career stage	Experienced
Location group	Texas (outside DFW)
Major (archive tag)	Computer Science
Sector area	Electronics & semiconductor manufacturing
Industry (LinkedIn)	Semiconductor Manufacturing
Pay listed	Not listed
Stated rules	No GPA or sponsorship rule stated
Posted / closes	2026-10-15 · 2026-12-20
Status when checked	Accepting applications on LinkedIn (Nov 1, 2026)
Source page	https://www.linkedin.com/jobs/view/4429622055/
Captured	2026-11-01

Full posting text as it appeared on LinkedIn on Nov 1, 2026, saved for ECS 1100. Postings change and close often; open the source page above for the current version before applying.

At a glance

What you would do

Engineer developing dispatching algorithms, dashboards, and automation software that schedule semiconductor fab production, and maintaining the servers they run on.

Who can apply

- **Majors named:** Computer Science, Industrial Engineering, Computer Engineering, Electrical Engineering
- **Degree and experience:** Bachelor's or above in CS, industrial, computer, or electrical engineering; strong programming plus expertise in areas like optimization or analytics.

Skills mentioned

Teamwork & collaboration · Communication · Problem solving & root cause · Automation & controls · Data analysis & statistics · Programming

Full posting text

A career at TSMC Arizona offers an unparalleled opportunity to contribute to the most advanced semiconductor fabrication facility in the United States. TSMC Arizona's first fab began production with leading-edge N4 process technology in the first half of 2025. The second and third fabs, utilizing even more advanced process technology, will start production by 2028-2030. America's leading technology companies will rely on TSMC Arizona for the next generations of chips that will power the digital future.

Job Summary

As a Dispatching Algorithm/Infrastructure Maintainer Developer, you will design, develop, and optimize complex algorithms to efficiently manage/support manufacturing and allocate resources, tasks, or routes within various systems. CIM engineers aim to automate factory production, minimizing human intervention. This involves developing dashboards,

assigning production tasks, and implementing automation solutions. They utilize diverse tools, computer technologies, and algorithms to streamline operations and enhance efficiency.

Essential Functions

- Server maintenance and related setting supports, including routine upgrade, troubleshooting with AP teams and virtual machines solutions etc.
- Coordinate the project for system enhancement and communicate requirement with users.
- Support application migration and participate in technical training from headquarters.

Required Qualifications

- Bachelor’s degree or above in Computer Science, Industrial Engineering, Computer Engineering, Electrical Engineering, or a related technical discipline.
- Strong programming and software engineering skills, including experience with data pipelines, APIs, automated testing, and performance optimization.
- Strong applied expertise in one or more areas such as computer vision, anomaly detection, predictive analytics, optimization algorithm, or scheduling, with the ability to guide work across adjacent domains.

Preferred Qualifications

- Familiarity in programming languages such as C/C++/C#, Java, Python, HTML5, SQL, VBA.Excellent collaboration & communication skills, both verbal and written, across different levels and functions of organizationExperience and skill in operation algorithms and A.I. is a plus.
- Experience and skill in system infrastructure architecture is a plus e.g., VMWare, Linux/MS solution, Nginx, K8S/Docker etc.
- Proficient in Microsoft Office Suite with ability to use a computer, keyboard mouse and other standard office equipmentStrong interpersonal skills to establish trust and maintain productive working relationshipsTeam-orientation and growth mindset to take on new responsibilities and share workload

Physical Requirements: [JJ2]

- Primarily an office-based role requiring the ability to sit for extended periods of time while working at a computer.
- Ability to use a computer, keyboard mouse and other standard office equipment for prolonged periods of time.
- Ability to occasionally stand, walk, reach, bend, and carry light objects.

Application Consideration Consent

By applying to this position, I acknowledge and agree that my application and qualifications may be considered for other roles within the company. However, I reserve the right to withdraw my consent for future consideration of other roles within the company at any time.

Benefits

As a valued member of the TSMC family, we place a significant focus on your health and well-being. When you are at your best physically, mentally, and financially, our company thrives. TSMC offers a comprehensive and competitive benefits program that includes:

- Medical, Dental, and Vision Plans: Choose the options that best fit your and your family’s needs.
- Income-Protection Programs: Financial assistance during injury or illness.401(k) Retirement Savings Plan: Secure your financial future with competitive employer contributions.
- Paid Time-Off Programs and Holidays: Recharge and spend quality time with loved ones.

TSMC is an equal opportunity employer. We celebrate diversity and are committed to creating an inclusive environment for all employees. All qualified applicants will receive consideration for employment without regard to race, color, religion, age, sex, sexual orientation, gender identity, national origin, disability, veteran status, or any other protected characteristic.

We encourage all qualified individuals to apply and welcome applications from diverse backgrounds and experiences. Candidates must be able to perform the essential functions of the job with or without reasonable accommodation. If you need an accommodation as part of the application process, please contact P_LOA@tsmc.com .